SCIENCE Y10 **Name:_____**

WORKSHEET ONE **TYPES AND TRANSFER OF ENERGY**

$E_p = mgh$ $E_k = \frac{1}{2}mv^2$

1. What is the unit for energy? _____ What is the symbol for this unit? _____(A)
2. List the names of six types of energy. (A) _____

   _____
3. Name the main type(s) of energy present in the following situations. (A)

   (a) An unlit match. _____

   (b) A stretched rubber band _____

   (c) An orbiting satellite _____

   (d) A thunder clap _____

   (e) A bolt of lightning _____

   (f) A kicked soccer ball _____
4. The sun is a giant hydrogen bomb. It uses _____energy to create solar energy. This solar energy is a mixture of different types of radiant energy such as _____ and _____. This radiant energy can travel through _____to earth and the earth absorbs some of it and _____some of it. Plants use chlorophyll to absorb light energy except for the colour _____ which they reflect. They store this energy in their leaves and roots as _____ potential energy. (A)
5. Write energy transfer equations for the following situations. (A,M)

   (a) An electric bell is switched on.

   _____ → _____ → _____

   (b) A bullet is fired up into the air.

   _____ → _____ → _____

   (c) A pine-cone falls off the top of a pine tree.

   _____ → _____
6. Usually when energy is transferred from one type to _____, some of the energy is also turned into _____. In a car motor, less than 50% of the chemical _____energy stored in the petrol is turned into the_____energy of

the moving car. The car engine gets very _____ and this unusable heat must be removed. The car radiator is filled with _____ which circulates through the motor and removes the heat. The hot water then circulates through the radiator and is _____ by the air passing over it. (A)

7. In each of the following pairs, one member of the pair has more energy. Which is it? (A)
(a) A kg of lead or a kg of feathers sitting on the same shelf. _____
(b) A 50cent piece on a table or a 50cent piece on the floor beside it? _____
(c) A cricket-ball hit by Noddy or a cricket-ball hit by Superman?? _____

8. Calculate the gravitational potential energy in a 1.6 kg pot-plant which is on a shelf 1.4 m above the floor. (Use $g = 10\ Nkg^{-1}$) (M)

9. Calculate the gravitational potential energy in a 0.4 kg seagull which is flying at a height of 200 m above the ground. (Use $g = 10\ Nkg^{-1}$) (M)

10. Calculate what height a 3kg object must have above the ground if it has 540 J of gravitational potential energy? (Use $g = 10\ Nkg^{-1}$) (M)

11. A boy and his bike have a combined mass of 70 kg. If the boy is cycling at 15 ms^{-1} how much kinetic energy is present? (M)

12. A bullet with mass 15g is fired at 200ms^{-1}. How much kinetic energy does it have? (M)

13. A 200g orange falls off a tree and just before it hits the ground it reaches a speed of 10 ms^{-1}.
(a) Calculate how much kinetic energy it has. (M)

(b) Calculate how high it was growing on the tree before it fell off. (E)